

Text Classification of Customer Feedback Using a Neural Network

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Abstract—This paper reports a no-code text-classification experiment in which a neural network is trained to sort customer-feedback messages into three intent categories: praise, complaint, and urgent support. Using the Machine Learning for Kids platform and a food-delivery business domain, a labeled dataset was collected, a model was trained, and its behavior was evaluated on ten previously unseen phrases. After an initial low-confidence observation, the dataset was refined to reduce overlap between the complaint and urgent classes. The refined model classified all ten clean test phrases correctly with an average confidence of 94.9 percent, while sarcastic and out-of-vocabulary phrases exposed its limits through markedly lower confidence. The paper discusses why tagged learning generalizes better than hand-coded rules, how dataset quality and volume govern accuracy, and where feature extraction occurs in the classification pipeline.

Keywords—neural networks; text classification; natural language processing; supervised learning; feature extraction; customer sentiment.

I. INTRODUCTION

Businesses receive large volumes of customer feedback every day, and sorting that feedback quickly is important for good service. A traditional way to automate this sorting is to write explicit rules, for example, telling the program that any message containing the word “refund” is a complaint. This approach becomes fragile as soon as language varies. The goal of this exercise is to build a classifier that learns from examples instead of rules, and to observe how it behaves when it meets new and unfamiliar text.

A neural network learns by adjusting internal weights so that patterns in tagged examples become associated with the correct labels. Rather than memorizing keywords, it maps the statistical features of the text to categories. This report documents the construction of such a model, presents its measured accuracy, and reflects on the questions raised by the exercise.

II. METHODOLOGY

A. Platform and Project Setup

The model was built on the Machine Learning for Kids text project, a no-code environment that exposes three stages: Train, where tagged examples are collected; Learn & Test, where the neural network is trained; and Make, where the model can be used in an application. A new text-recognition project was created and three classes were defined: Positive_Praise, Negative_Complaint, and Urgent_Support.

B. Dataset and Domain

A food-delivery application was chosen as the business domain because its customer feedback naturally spans all three intents. The dataset was built in two rounds: an initial

small, partly overlapping set, then an expanded set of ten distinct example phrases per class, giving a balanced thirty-example training set. The examples were written to reflect realistic customer language, such as praise for fast delivery, complaints about cold food, and urgent messages about double charges or locked accounts.

C. Training

During the Learn & Test stage, the platform converted each phrase into numerical tokens and trained a neural network to associate those token patterns with the class labels. Once training completed, the model status was reported as available, confirming that the classifier was ready for inference.

D. Testing and Refinement

A test set of ten new phrases, none of which appeared in training, was prepared with a mix of clearly positive, clearly negative, and clearly urgent messages. An early version of the model classified an ambiguous complaint as urgent with only 44 percent confidence, revealing overlap between the two negative-toned classes. The dataset was then refined and expanded to ten examples per class so that complaint and urgent examples were more distinct, and the model was retrained. In addition to the ten clean phrases, four adversarial phrases were prepared to probe failure modes: two sarcastic messages that carry positive words but negative meaning (Test Case B), and two out-of-vocabulary administrative queries for which no class exists (Test Case C).

III. RESULTS

Table I lists every test phrase together with the class the model predicted and its confidence score. Table II summarizes the accuracy computed for each category.

TABLE I. TEST RESULTS ON TEN UNSEEN PHRASES

#	Test Phrase (unseen)	Exp.	Pred.	Conf.	OK
1	The food was tasty and arrived very quickly!	Pos.	Pos.	93%	Y
2	I love how simple and smooth this app is.	Pos.	Pos.	96%	Y
3	Fast delivery and the food is always amazing.	Pos.	Pos.	93%	Y
4	My order never arrived and support ignored me.	Neg.	Neg.	94%	Y
5	Cold food again, such a waste of money.	Neg.	Neg.	100%	Y
6	Waited two hours and got the completely wrong order.	Neg.	Neg.	96%	Y
7	I was charged twice, please fix this immediately!	Urg.	Urg.	95%	Y
8	My account was hacked, I need urgent help now!	Urg.	Urg.	96%	Y
9	Payment taken but no order, respond right away!	Urg.	Urg.	89%	Y

10	Locked out of my account and charged twice, help now!	Urg.	Urg.	97%	Y
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a. Pos. = Positive_Praise, Neg. = Negative_Complaint, Urg. = Urgent_Support.

TABLE II. ACCURACY PER CATEGORY

Category	Correct	Total	Accuracy	Avg Conf.
Positive_Praise	3	3	100%	94%
Negative_Complaint	3	3	100%	97%
Urgent_Support	4	4	100%	94%
OVERALL	10	10	100%	94.9%

After refinement, the model classified all ten unseen phrases correctly, giving an accuracy of 100 percent for each of the three categories and 100 percent overall. Confidence ranged from 89 to 100 percent, with an average of 94.9 percent. The improvement from the earlier 44 percent result confirms that clearer, better-separated examples raise both accuracy and confidence.

A. Failure-Mode and Edge-Case Behaviour

To probe the model's limits, four adversarial phrases were tested (Table III). The two sarcastic messages were still classified correctly as complaints, but their confidence fell to 67 and 82 percent, well below the 89 to 100 percent seen on clear phrases. This shows that positive words such as "great" and "fantastic" pull the decision toward praise and leave the model less certain. The two out-of-vocabulary administrative queries, which belong to none of the three classes, produced confidence of only 4 and 5 percent, revealing that the model is effectively guessing when it meets an intent it was never trained on. Both weaknesses have clear fixes: adding explicitly sarcastic examples to the complaint class would raise the model's confidence on ironic wording, and creating a dedicated General Inquiry class would give administrative questions a proper home instead of forcing a low-confidence guess.

TABLE III. FAILURE-MODE AND EDGE-CASE RESULTS

Test Phrase (adversarial)	Type	Pred.	Conf.
Oh great, another crash right in the middle of my order.	Sarcastic	Neg.	67%
Fantastic, waited two hours for cold food again.	Sarcastic	Neg.	82%
Can I change my billing address?	Out-of-vocab	Urg.	4%
What are your customer service hours?	Out-of-vocab	Neg.	5%

IV. DISCUSSION

A. Explicit Rules versus Tagged Learning

Writing explicit IF/ELSE statements for every possible complaint runs into several problems. Customers express the same idea in countless ways, so no finite list of keywords can cover all of them; a rule that flags "refund" would miss "I want my money back." Rules also collide with one another, because a single message can contain words that belong to different categories, and the programmer must then decide priorities by hand. Maintenance grows heavier over time, since every new phrasing, product, or slang term requires another rule. Finally, rules ignore context: the word "great" is positive in praise but negative in sarcasm, and a fixed rule cannot tell the difference. Tagged learning avoids these problems by inferring patterns from examples, so the model can generalize to wordings it was never explicitly told about.

B. Data Bias and Dataset Quality

The exercise showed a clear link between data and performance. With only about three to five overlapping examples per class, the model produced a weak 44 percent prediction on an ambiguous phrase, because it had too little evidence to separate a complaint from an urgent issue. After the dataset was expanded to ten distinct and representative examples per class, the same style of phrase was classified correctly with confidence above 90 percent. This direct three-versus-ten comparison shows how sharply added data lifts both accuracy and certainty. Deep learning relies heavily on data quality, volume, and diversity because the network has no outside knowledge; everything it "knows" comes from the examples it is given. Too few examples leave gaps the model must guess across, while biased or repetitive examples teach narrow patterns that fail on real, varied input. A larger and more diverse dataset gives the network more evidence, producing decisions that are both more accurate and more confident.

C. Locating Feature Extraction in the Pipeline

Feature extraction is the step that turns raw text into the numerical values a neural network can process. In this pipeline it happens at the start of the Learn & Test stage, before the network updates any weights. My best estimate is that the platform first tokenizes each phrase, splitting it into words or sub-words, and then converts those tokens into numerical vectors, for example counts or embeddings, that represent the presence and pattern of words. These vectors are the features. Only after this conversion does the network map the features to the class labels during training. In other words, feature extraction sits between the tagged text input and the weight-learning process, transforming language into the mathematical form the classifier actually learns from.

V. CONCLUSION

This exercise demonstrated that a neural network can classify customer feedback accurately without any hand-written rules. Starting from a balanced food-delivery dataset, refining it to remove class overlap, and retraining produced a model that classified all ten unseen phrases correctly with an average confidence of 94.9 percent. The results reinforce three lessons: tagged learning generalizes where explicit rules break down, dataset quality and diversity drive performance, and feature extraction is the pivotal step that converts text into learnable features. Future work could add more classes, larger datasets, and deliberately sarcastic examples to further harden the model against ambiguous language.

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